

Data Visualization Final Report

1. Introduction

In the past several years, AI has shifted from an emerging technology to an advanced mainstream tool reshaping the job market for computer science professionals. For us, as senior computer science students, we're on the verge of entering this market, and the implications of that shift are what we'll be facing after graduation. This project explores the CS job market through the lens of AI and how it serves as a potential threat to future job opportunities. The data we sourced are from the U.S. Bureau of Labor Statistics, Kaggle, and Mendeley Data. These data hold valuable information that helps us analyze factors such as adoption of AI, automation risks across industries, trends for wages and compensation, and geographical distribution of the tech industry. Our goal is to present what AI imposes upon current and upcoming computer scientists.

Data Sources:

- AndrewMVD. *Occupation, Salary, and Likelihood of Automation*. Kaggle Dataset.
- Biswas, Richmond Antor Biswas; Faria Jahan, Janie (2026), "Global AI Adoption Trends by Industry Sector, Country, and Business Function (2017–2025): A Compiled Research Dataset," Mendeley Data, V2, doi: 10.17632/dnm5jxgn2m.2
- U.S. Bureau of Labor Statistics. "Occupational Employment and Wage Statistics Profile." *U.S. Department of Labor*, 2025, data.bls.gov/oesprofile/.
- U.S. Bureau of Labor Statistics. "2024 Annual Averages – Household Data – Tables from Employment and Earnings." *Current Population Survey*, U.S. Department of Labor, 20 Feb. 2026, www.bls.gov/cps/cps_aa2024.htm.
- U.S. Bureau of Labor Statistics. (2026). *Table 25b. Unemployment levels and rates by detailed occupation, 2025 annual averages*. Current Population Survey. March 10, 2026, <https://www.bls.gov/cps/cpsaat25b.htm>

- Biswas, R. A., & Jahan, J. F. (2026). *Global AI Adoption Trends by Industry Sector, Country, and Business Function (2017–2025)* (Version 2) [Data set]. Mendeley Data. <https://doi.org/10.17632/dnm5jxgn2m.2>
- National Center for Science and Engineering Statistics. (2025). *National Survey of College Graduates (NSCG)* [Public use microdata]. U.S. National Science Foundation & U.S. Census Bureau. <https://nces.nsf.gov/explore-data/microdata/national-survey-college-graduates>

2. Methodology/Data Preparation

For the “Global AI Adoption Trends by Industry Sector, Country, and Business Function (2017–2025): A Compiled Research Dataset,” the first simple transformation process was converting all decimal values into proper percentages. Next, filtering data, we narrowed down the sectors to exactly 8 sectors including technology, finance, healthcare, retail, manufacturing, education, government, and agriculture, to reduce visual clutter. Additionally, rows where adoption values for 2022 or 2025 were dropped to ensure complete data points. Next, transforming data, we condensed the sector names to be shorter to reduce visual clutter. Additionally, we refactored the data from a wide format to a long format. Lastly, sorting the sectors by their rate in descending order ensured charts were intuitive and easy to understand.

For the Kaggle “Occupation, Salary, and Likelihood of Automation” dataset, it was downloaded directly from Kaggle rather than being downloaded as a CSV file. It contains two files within it: a spreadsheet and a CSV file. The CSV required text encoding during the loading implementation because of special characters in the data. Next, joining/merging data, the two files shared a standard occupation classification code, just labeled differently, so the codes were used as a key to join the two tables together. An inner join was specifically used, so we narrowed down to only occupations that appeared in both files. After merging, we removed duplicate occupation names that could’ve carried over due to redundancy. Next, filtering data, we extracted the subsets of interest, so the ten occupations with the highest and lowest

automation risk were in different tables to highlight the extremes on both sides. Lastly, transforming data, generalized job titles, and risk values were formatted as percentages.

For the “U.S. Bureau of Labor Statistics. (2026). *Table 25b. Unemployment levels and rates by detailed occupation, 2025 annual averages*” dataset, it was downloaded from the Bureau of Labor Statistics website as an Excel (.xlsx) file. The Excel file contained header metadata in the first 3 rows. The dataset was then loaded into Collab using the Pandas library, skipping the initial rows with header data. To focus on computer science–related occupations, a subset of relevant occupations was defined after viewing the spreadsheet locally. The occupations that we chose to narrow down to are: software developers, software quality assurance analysts and testers, web developers, web and digital interface designers, and ‘other computer-related’ roles. The data frame was filtered to retain only these CS-related occupations, and a new column for the year 2025 was added to facilitate time-series analyses and merging with other datasets. Finally, the cleaned and filtered data was exported as a brand new CSV file.

For the “National Center for Science and Engineering Statistics. (2025). National Survey of College Graduates (NSCG) [Public use microdata]” dataset, it was downloaded from the U.S. National Science Foundation and U.S. Census Bureau website as a CSV file. The dataset contains individual-level survey responses including labor force status, survey weights, and major codes. To focus on computer science–related fields, a subset of relevant major codes was defined, and the data frame was filtered to retain only sampled individuals with CS-related first bachelor’s degrees. Weighted estimates were computed to derive employment, unemployment, and labor force participation rates. The filtered dataset was then cleaned to include human-readable column names, a mapped labor force status label, and a year column for 2023, and exported as a CSV file named `cs_related_employment_2023.csv` for further analysis.

For the “Biswas, R. A., & Jahan, J. F. (2026). Global AI Adoption Trends by Industry Sector, Country, and Business Function (2017–2025) (Version 2) [Data set]” dataset, it was downloaded from Mendeley Data as an Excel (.xlsx) file. The original file contained header metadata in the

first three rows and included multiple time entries for some years, such as “Early 2024” and “Late 2024.” The dataset was loaded into Python using pandas, with the initial metadata rows skipped. To create a consistent yearly time series, these split-year entries were merged, and the AI adoption rates were averaged for the combined year. The adoption percentage column was converted to numeric values, and the dataset was aggregated by year to produce a clean time-series of AI adoption rates. Finally, the cleaned data, containing only the year and AI adoption percentage, was exported as a CSV file named ‘AI_Global_Trends_Dataset.csv’ for further analysis.

3. Results (Plots)

3.1 AI Adoption

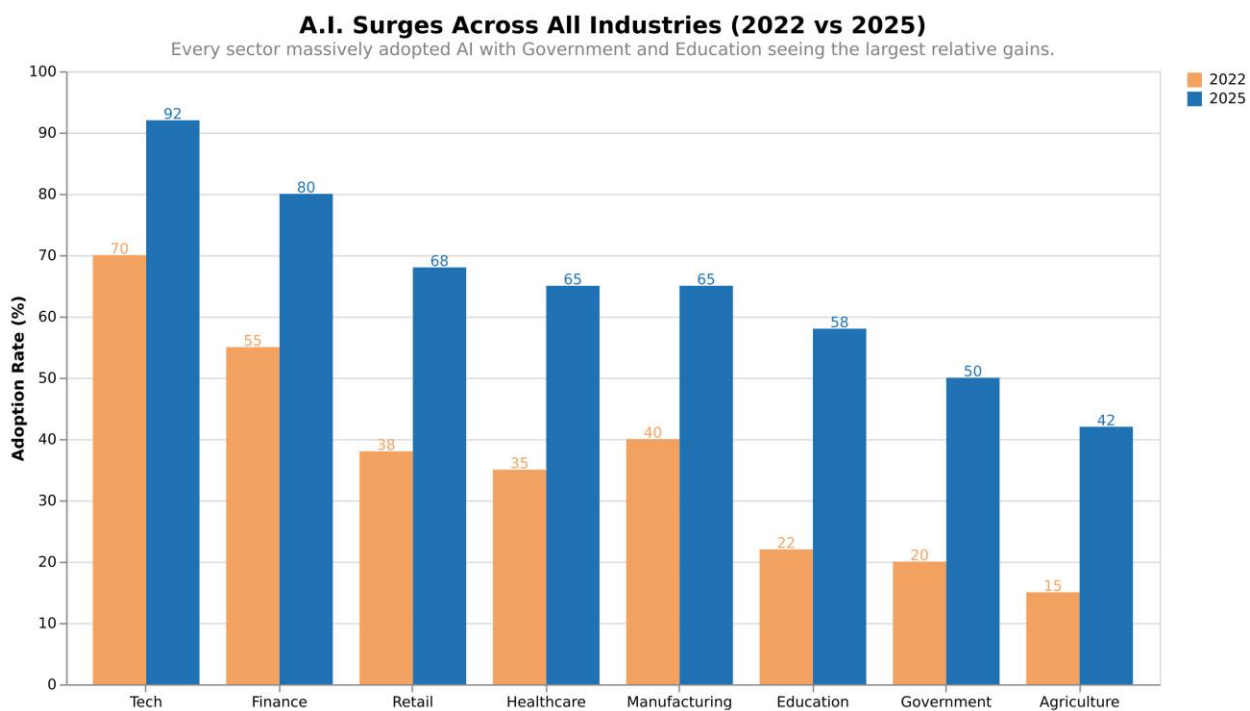


Figure 1. Bar chart showing AI adoption rates by 8 industries, 2022 vs. 2025.

3.2 Automation Risk

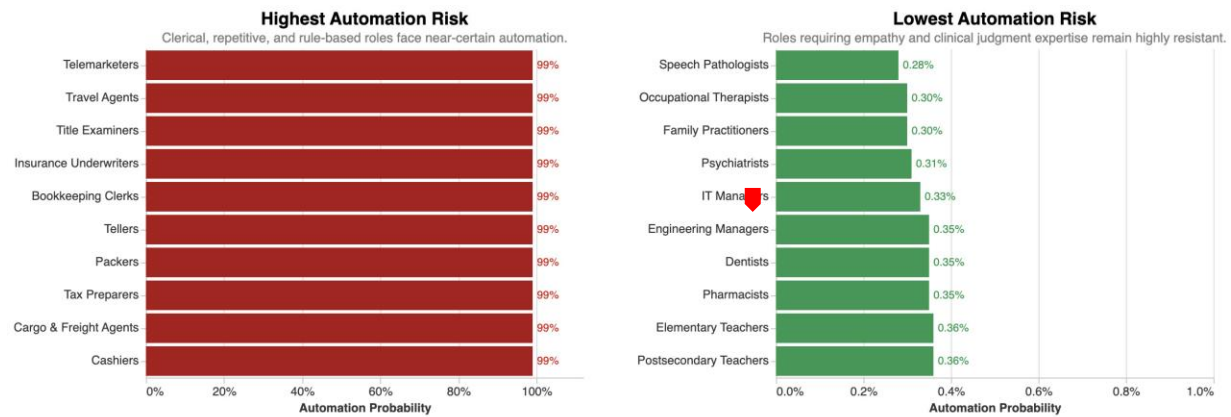


Figure 2. Dual bar chart comparing the top 10 occupations with the highest and lowest automation risk

3.3 Growth of Investment for AI

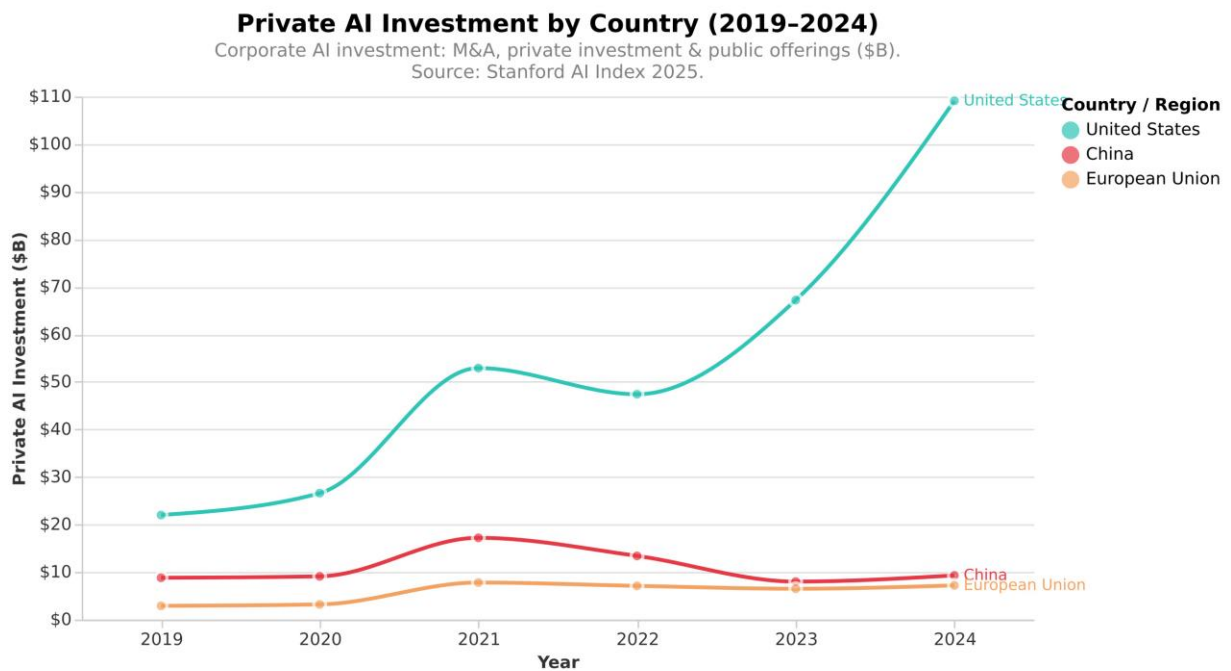


Figure 3. Line chart showing private AI investment by 3 countries (2019-2024).

3.4 Employment and Company trends

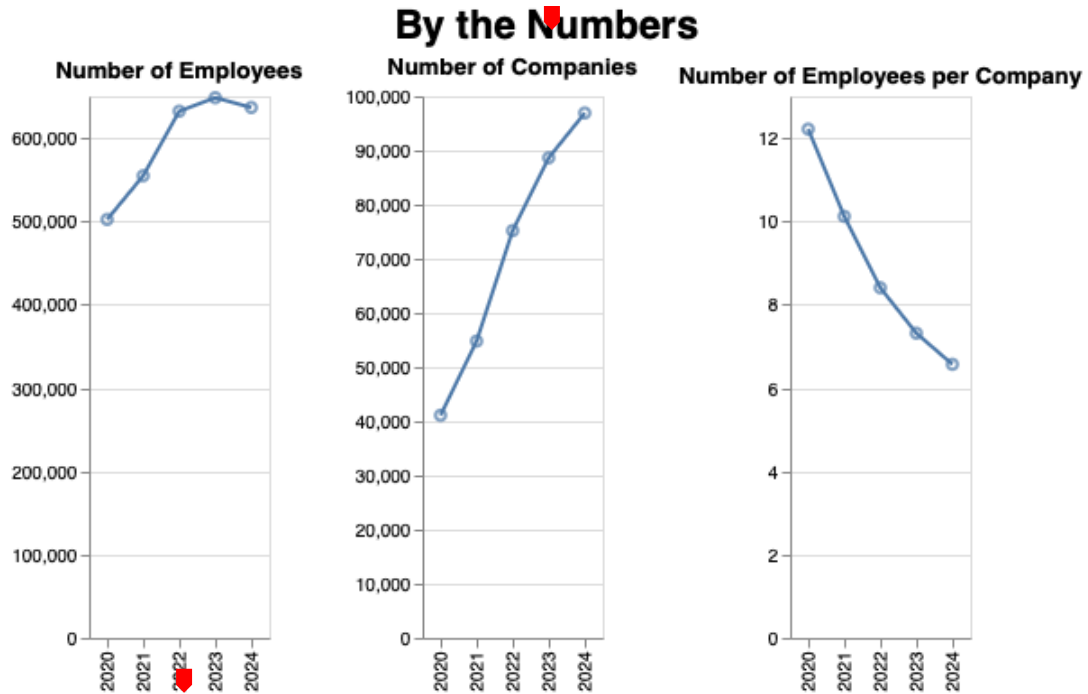


Figure 4. 3 Line charts showing the Number of Employees, Companies and Employees per Company (2020-2024).

3.5 Monetary Analysis of National Annual Wages and Employee Salaries

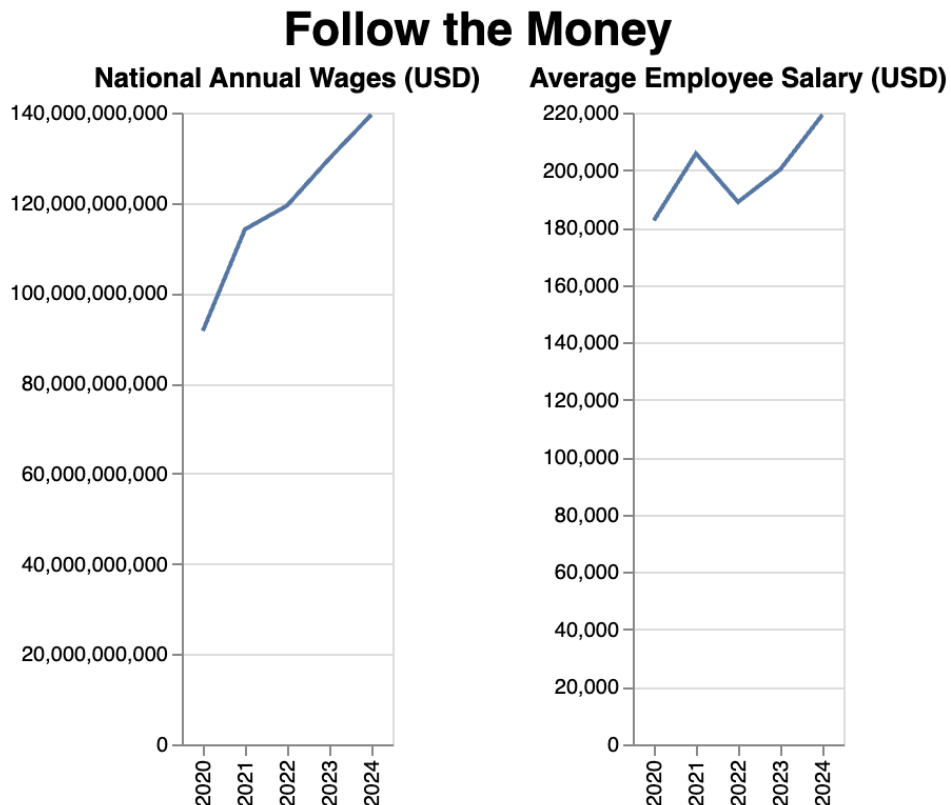


Figure 5. 2 Line charts showing the National Annual Wages and Employee Salaries (2020-2024).

3.6 National Map showing the Number of Companies per state (2024)

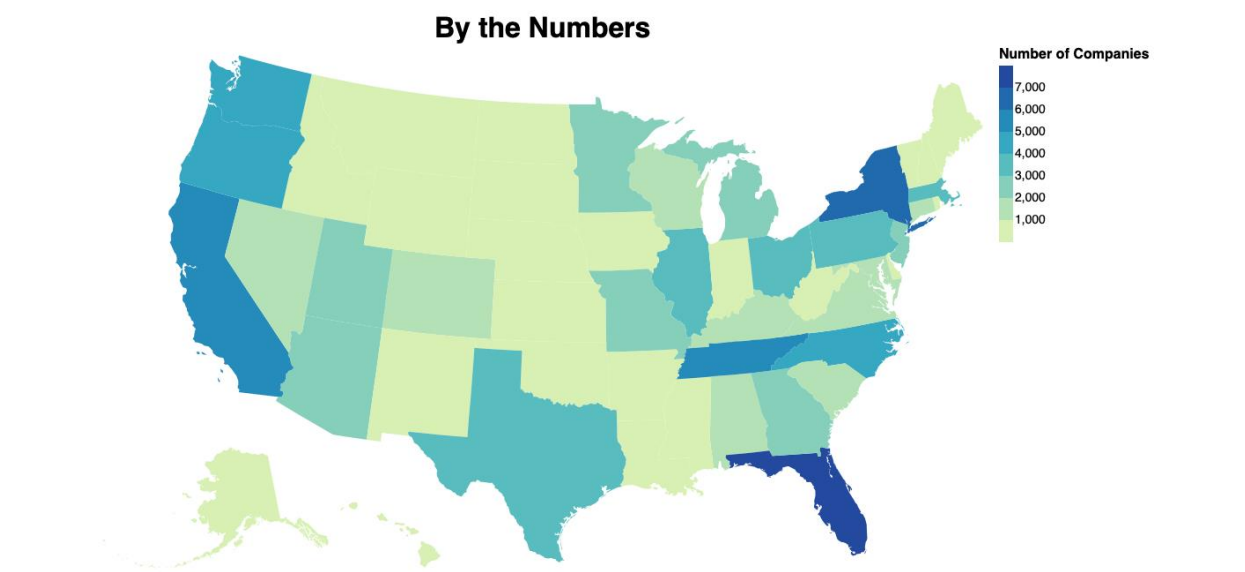


Figure 6. A Choropleth displaying the number of companies in each state (2024).

3.7 Monetary Analysis of Combined National Wages per State

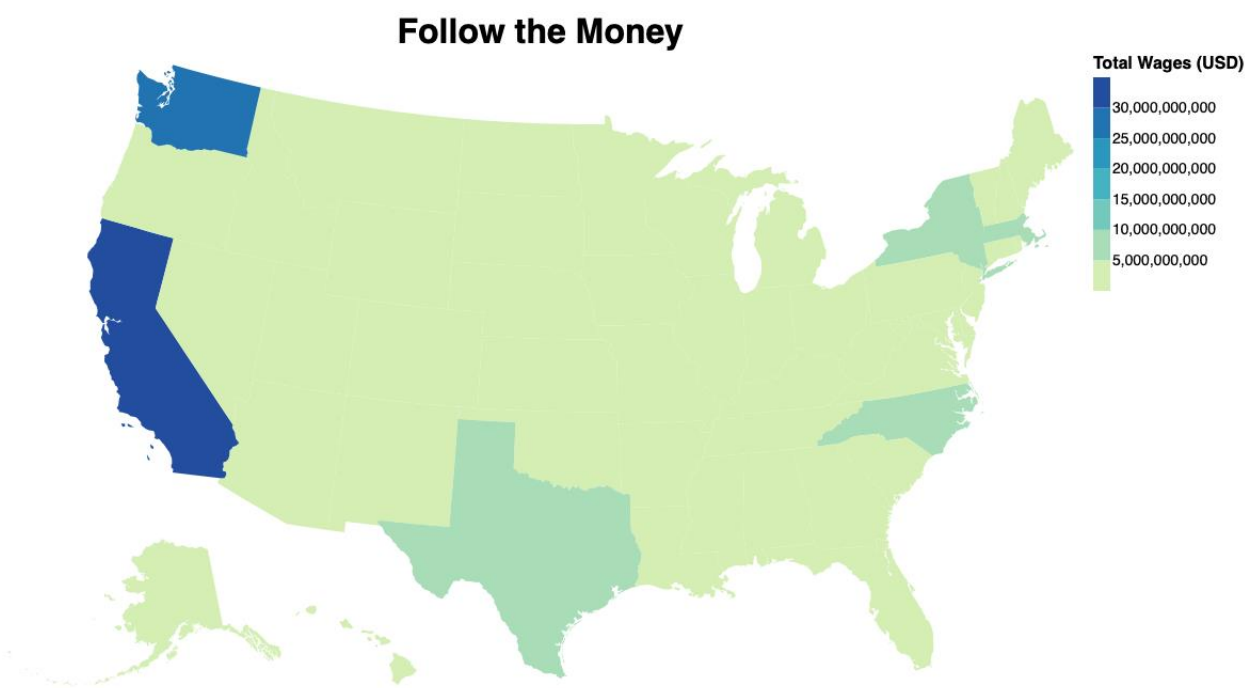


Figure 7. Total Employee Compensation paid in each State (2024).

3.8 Base Salary vs. Total Compensation by Gender and Sector

The Hidden Compensation

○ Base Salary ● True Total Compensation

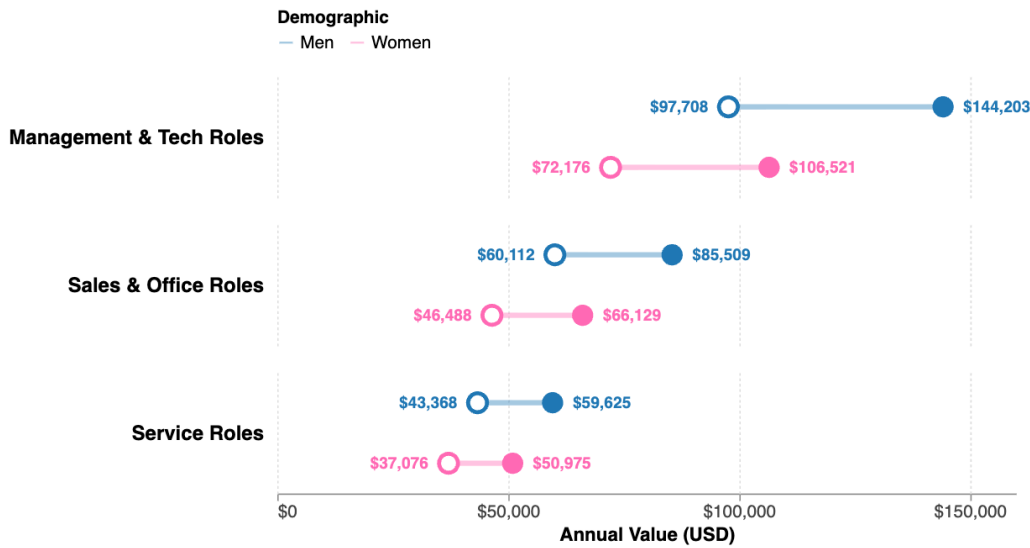


Figure 8: Horizontal dumbbell chart to illustrate widening gap across between base salary and total compensation for men and women across different occupational sectors.

3.9 Proportion of Compensation Components in Private vs. Public Sectors

The Public Sector Equalizer: Swapping Base Pay for Benefits

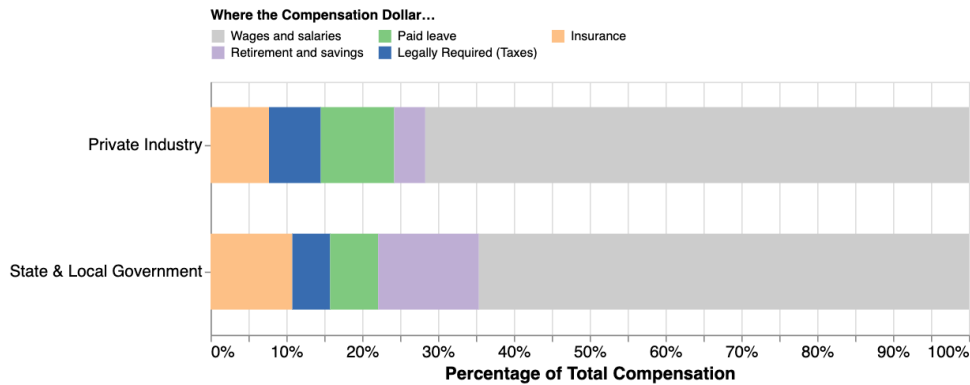


Figure 9: Normalized stacked bar chart to compare the allocation of compensation dollars between private industry and state / local government roles.

3.10 Total Employment by Gender in High Benefit vs Low Benefit Sectors

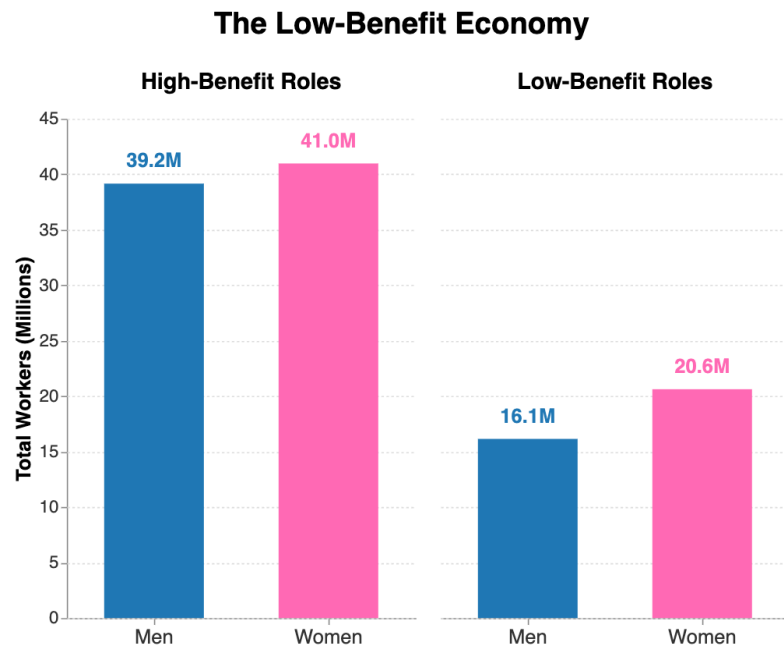


Figure 10: Presents a faceted bar chart comparing total number of male and female workers concentrated in high vs low benefit economic sectors

3.11 Union Pay Premium by Demographic Group

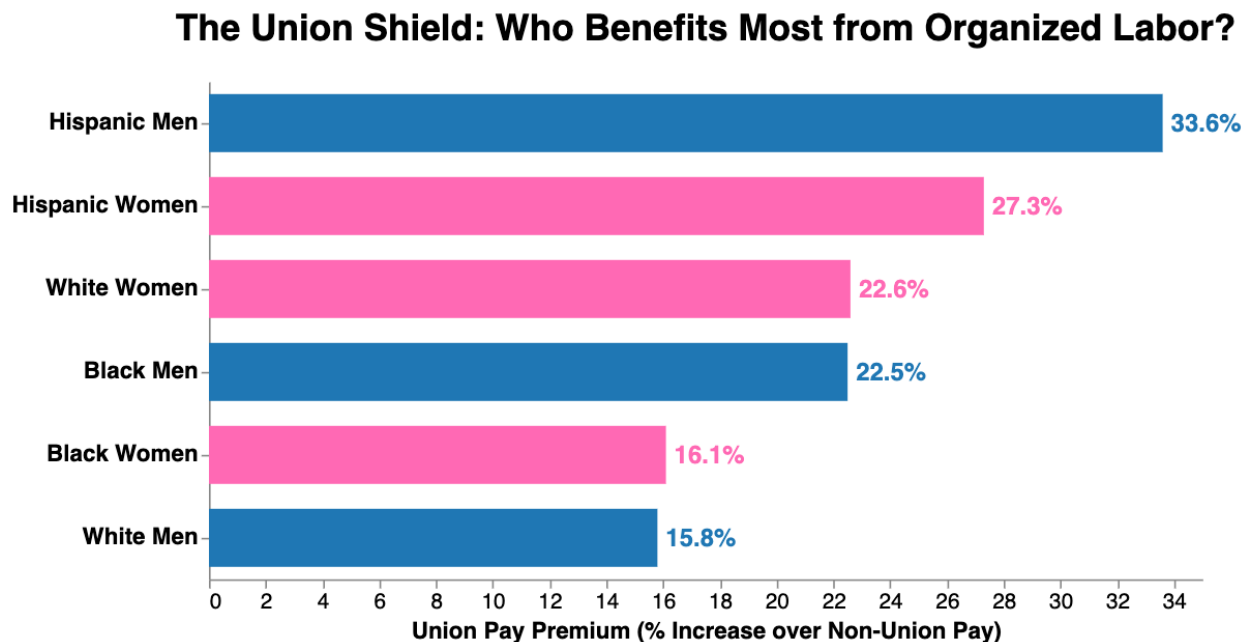
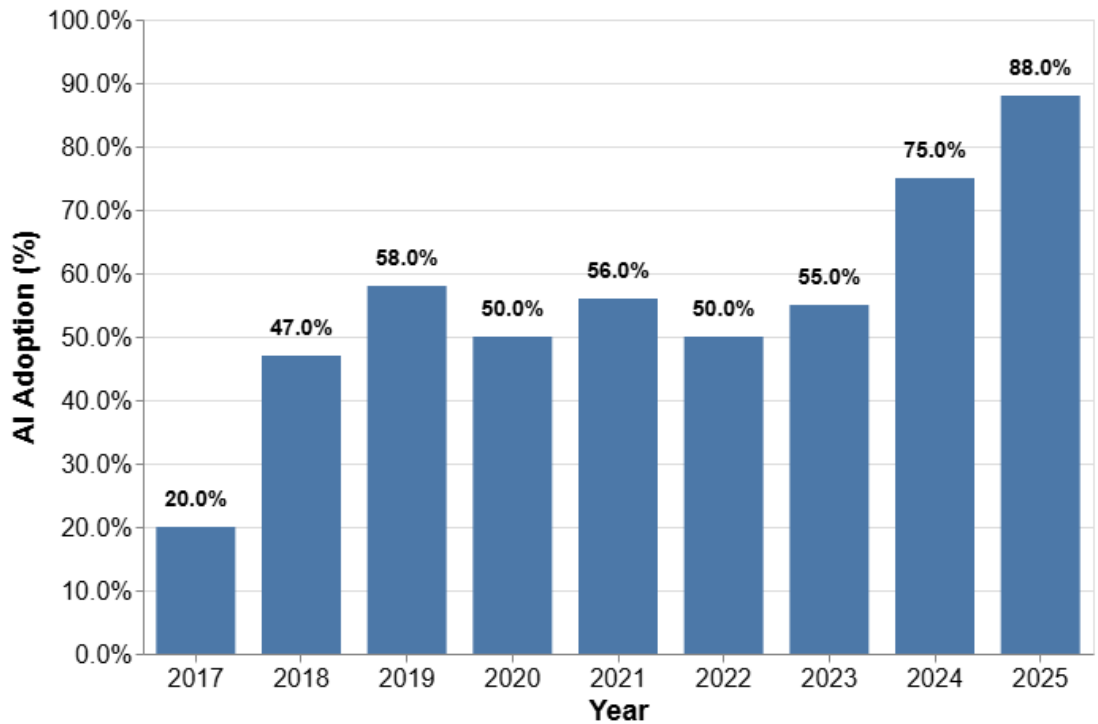


Figure 11: Horizontal bar chart to display the percentage increase in weekly earnings provided by union affiliation across different racial and gender demographics

3.12 AI Adoption over Time (2017-2015)

AI Adoption Over Time (2017-2025)

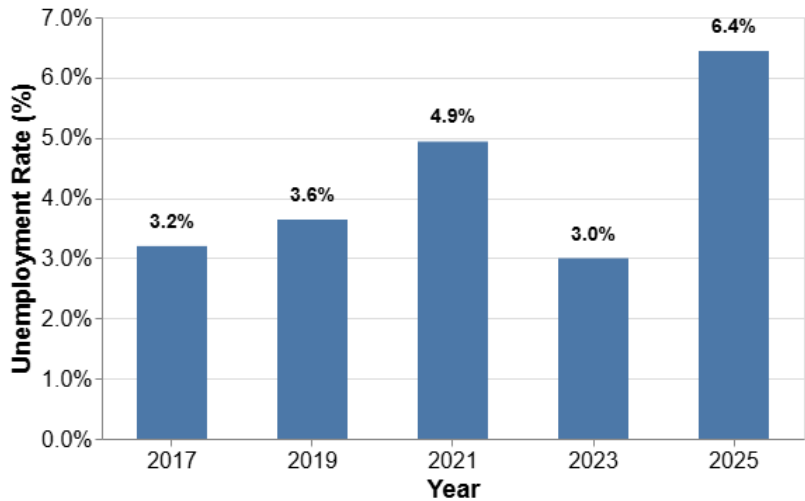
AI Adoption steadily increases from 2017-2025, with the sharpest increase seen in 2022-2025 (from 50% up to 88%).



3.13 Unemployment Rate for Computer Science-Related Graduates (2017-2025)”

Unemployment Rate for Computer Science-Related Graduates (2017–2025)

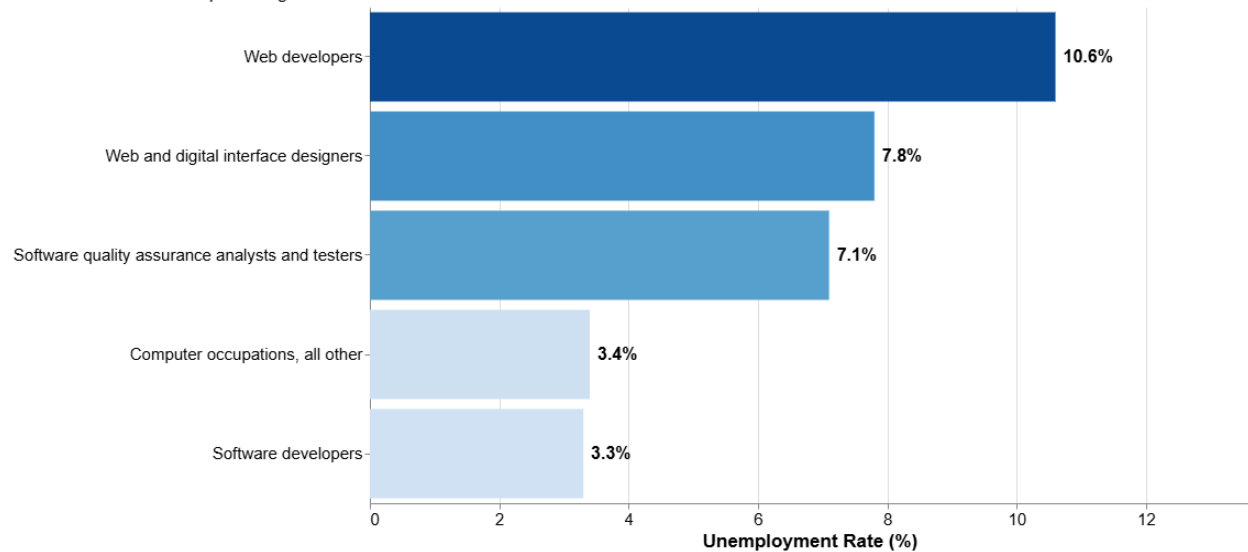
Unemployment remains at around the 3-4% level from 2017-2019, then spikes in the 2020s, with large spikes to 4.9% in 2021 and 6.4% in 2025.



3.14 Computer Science Occupation Unemployment Rates (2025)

CS Occupation Unemployment Rates (2025)

Detailed breakdown with percentage labels and AI-risk context



4. Design Justification

3.1.

The bar chart was chosen as the encoding mark because we're comparing quantitative data (AI adoption rate) across two categorical fields (industry sector and year), making bar length the most effective encoding channel for comparing the different magnitudes across the 8 different sectors. We grouped bars with xOffset because, instead of stacking the different yearly bars, we placed them side by side. Stacking would misinform viewers about what's intended. Grouped bars preserve the comparison factor. Color encoding helps differentiate which data is associated with which year. The color we use is conventional and helps contrast. Additionally, it's accessible for color-blindness. Color overall reduces the cognitive load needed to differentiate. Data labeling was used to reduce eye movement for viewers, which means assigning labels or categories to these data points, making information easier to process visually. Additionally, the label color mirrors the bar colors, keeping the chart consistent without a lot of visual clutter. Scale the y-position domain; the scale is anchored to the full

percentage range of 0-100%. Lastly, the X-position label angle is set to 0 since horizontal labels are more readable than vertical ones.

3.2

The horizontal bar chart was chosen as the mark encoding because we're comparing a quantitative data type (automation probability/risk) across a single categorical field (occupation), making bar length an effective encoding channel for highlighting differences in magnitude across the occupations. Additionally, the orientation of the bars being horizontal was deliberate, allowing texts to be legible and emphasizing the comparisons. The side-by-side layout was chosen to separate and highlight the top 10 highest-risk and lowest-risk occupations. Merging the two would cause misleading information and wouldn't convey our message. Also, the color scheme was used intentionally, as we associate these colors easily with understandable norms, for example, the highest risk is encoded in red to emphasize the danger, while the lowest risk is encoded in green to emphasize the safety. The y-position are sorted by their quantitative value, descending for the high-risk panel and ascending for the low-risk p panel, so they can emphasize their most extremes at the top. Data labels were added using `mark_text` to add a small data ink for their percentage, as it may be hard to interpret without it, also it eliminates eye movement and makes it sufficient and consistent. Lastly, the x-position is formatted as percentages using the format "%" rather than displaying the raw decimal probability.

3.3

The line chart was chosen as the mark encoding because the data displayed is a quantitative measure (private AI investment) tracked across years (2019-2024), making a line an effective encoding for showing change and trend over time. Lines help emphasize change between data points, which answers what we are trying to convey, which is how the investment trajectory evolved throughout the half-decade. Color encoding was applied to the country field to differentiate between the filtered countries (US, China, and European Union) across all encodings. Color is a natural channel for categorical separation. By applying the same color scale for all marks and encodings, it helps maintain visual consistency, reducing any visual

clutter or cognitive load. Data labels were added and placed using `mark_text` to the final year, eliminating the need for viewers to cross-reference between the color, points, and legend, reducing cognitive load. We set the `y-position scale to zero=false` so that we don't force inclusion of zero baseline, which means we don't obscure variations and trend differences since we want to communicate the relative change and trajectory across these countries over the years. Additionally, we formatted the values as dollar amounts to clearly communicate with the financial unit. Lastly, the x-position keeps the year value legible by making the `labelAngle=0`.

3.4

As the job market continues to develop, especially in the past year or so with AI, it is important to understand where the market has been in recent years. To showcase this data, 3 line plots using position, position, and subsequently position for the type of data plotted showcase this data in an easy-to-compare way, showing the trends of each variable employee and company. Though it can be seen that the companies are growing faster than the number of employees, the third chart shows this direct relationship, highlighting just how quickly the number of employees per company is dropping. The mark of a line and point shows exactly where the data points are as well as the trend of the data in a certain direction. These showcase the data in the easiest way to understand and visualize the trends effectively.

3.5

Money is an important factor in the job market, as it shows the financial investment into their workforce each year. For the computer science industry, it is important to analyze the growth of the industry and how that affects employee salaries and potential openings. The position. Position encoding using mark points and lines connecting them is beneficial to view and analyze the monetary effects of the industry and potential future.

3.6

To showcase where the locations of software companies are located a choropleth demonstrates the count of companies in each state, which is important, especially when location in the country is important to the new graduates trying to enter the job market and

determine where they would like to live. If determining whether to live on the east or west coast of the country, it is helpful to see which states have the most opportunity. As the map uses position-position encodings already, the next best encoding channel is color hue for the data. Since the quantitative data is binned, it gets treated as ordinal data.

3.7

This choropleth map is important to analyze the monetary investment that companies have in employee compensation for each state. This is also very helpful for developers entering the industry, as it shows where the most money is and where opportunities are more likely to be. To show this, the amount of data is binned to show the color encoding in a more accurate way. The position encoding channels are used on the map, which is helpful to show where in the country money is flowing geographically, thus the next highest encoding channel is color.

3.8

The horizontal dumbbell plot was selected to illustrate the differences between the hollow dot (Base Salary) and a larger solid dot to represent the ending point (True Total Compensation). The connecting lines provide a physical representation of the “Benefits Boost”. Direct data labels were added to each side of the dumbbell along with universally recognized color encodings (blue and pink) to differentiate between men and women, reducing the cognitive load for the audience.

3.9

The normalized stacked bar chart is the arguably the most effective way to compare proportions across different groups, regardless of their absolute totals. By normalizing the data to represent a single compensation dollar the audience can compare the structural differences in how private and state/ local governments pay their employees. **Distinct categorical colors** were applied to non-wage benefits to differentiate between benefit types (retirement, insurance, and paid leave).

3.10

The data was grouped into two massive, faceted buckets (high benefit & low benefit) to make the complex economic distribution instantly readable. A standard bar chart was chosen for its simplicity, and absolute worker counts were converted to millions on top of the bars to ensure universal readability. This reduced the cognitive load from audiences by effectively removing the need to trace gridlines back to the Y-axis.

3.11

A horizontal bar chart was used to accommodate the long demographic labels on the y-axis without overlapping the text. The data is explicitly sorted in descending order to guide the viewer toward demographics that are impacted the most. The chart intentionally retains the blue/pink color encodings to maintain visual consistency between men and women. Explicit percentage labels at the end of each bar provide immediate quantitative takeaways.

3.12

The vertical bar chart was chosen as the encoding mark because we are comparing quantitative data (AI adoption percentages) across a single categorical field (year), making bar length the most effective channel to convey relative magnitudes over time. Using vertical bars preserves chronological order and allows easy comparison of adoption growth year over year. Data labels are added directly above each bar to display the exact adoption percentage to help with viewers easily interpret key values. The Y-axis is scaled from 0 to 100% to anchor the data within the full possible range and provide context for growth. Font sizes for axis labels, titles, and labels are increased to enhance clarity, and the subtitle highlights key trends, helping viewers quickly grasp the most important insights. Overall, this design emphasizes accuracy, readability, and ease of comparison across years.

3.13

The vertical bar chart was selected as the encoding mark because it effectively shows changes in quantitative data (unemployment rate) across a time series (year), allowing viewers to easily track trends over time. Each bar represents the weighted unemployment rate for CS-related

graduates for a given year, making magnitude comparisons straightforward. A consistent color was used for all bars to maintain focus on height as the primary encoding channel, reducing unnecessary visual complexity. Data labels are placed directly above each bar to show exact percentages, improving interpretability. The Y-axis is scaled with a maximum slightly above the highest unemployment rate to provide visual headroom for the labels and ensure labels do not overlap outside the graph. Layering bars with text labels reduces cognitive load and emphasizes key values without cluttering the chart. Overall, this design highlights year-to-year trends in computer science unemployment clearly, and supports accurate comparisons between different time frames.

3.14

The horizontal bar chart was chosen as the encoding mark because we are comparing quantitative data (unemployment rates) across multiple categorical fields (computer science occupations), making bar length the most effective visual channel for comparing magnitudes from multiple categories. By using a horizontal orientation, longer occupation names are easier to read without rotation, improving accessibility and readability. Color encoding is applied to the bars using a blue sequential scheme to indicate intensity of unemployment, providing an intuitive visual cue. Percentage labels are added directly on the bars to reduce eye movement and cognitive load, allowing viewers to quickly see the exact values without needing to trace axis ticks. This is also for accessibility, especially for viewers who can't interpret colors well. The X-axis scale includes a small headroom above the maximum value to prevent label clipping, and both axis labels and titles are set with larger font sizes to improve clarity. Overall, this design balances accuracy, readability, and accessibility, making it straightforward to compare unemployment rates across CS occupations at a glance.

5. Analysis

3.1.

This chart illustrates the significant increase in AI adoption across all sectors from 2022 to 2025, emphasizing how quickly AI has grown from an emerging piece of technology to a global mainstream tool. In 2022, all sectors adoption rates were moderate, reflecting early stages of the AI boom. Tools from companies such as OpenAI and other companies integrating their own AI models began to gain traction, but still everything was experimental.

By 2025, every industry has begun adopting substantially more, suggesting AI has moved from something experimental to full-on use for implementations. However, it's notable that tech and finance continue to lead adoption rates, although they were already relatively high in 2022. This means that they advanced their existing AI integration, leading to greater efficiencies in their operations. Furthermore, agriculture being the outlier highlights limitations, and sectors that rely on manual processing or don't have the digital resources face barriers to adoption.

Overall, the chart illustrates not only how AI has surged in every industry but also how adoption patterns differ depending on the industry's readiness or infrastructure.

3.2

This chart illustrates how repetitive, rule-based roles dominate the high-risk panel. These roles share a simple common feature structurally: they can be reduced to a logical step that ML or robotic processing can handle more efficiently. Although 100% is not for certain, it indicates more so that technical barriers to these roles have fallen dramatically. So, it entails when these jobs will be replaced rather than if they will be. In contrast, low-risk panels are consistent with what it entails. These clusters of jobs near zero are due to the technical limitations, these roles are inseparable from human connection/relationship. Overall, the chart draws the idea that as automation risks intensify, a key takeaway we get is that relationship-, judgement-, or care commitment-intensive work will be to our advantage.

3.3

The chart demonstrates how the United States has significantly outperformed every other country, with the gap widening annually. A notable feature of this chart is the United States

line, its curve grew exponentially in 2022-2024, reaching investment levels that make the other countries look insignificant on the chart. The steepness of the 2023-2024 period underscores the post-OpenAI/ChatGPT investment surge that swept the financial market. Overall, this chart draws the interpretation that private AI investment isn't distributed evenly globally in the financial market. It's concentrating, the US is in favor of not just leading but fully dominating. In a sense, this chart isn't just a financial visualization but a map of where AI's future is being written.

3.4

The number of total employees in the software publishing sector is very important to analyze, as it shows how the job market is changing/trending. As computer science majors entering the sector, we are looking for opportunities to enter the workforce. Since the number of employees steadily rose from around 500,000 to just under 650,000 in 2023, it would seem as though the employment is growing steadily, but with a slight dip in the workforce in 2024, it can bring cause for concern as we enter the job market in 2026. If that slight dip turns into a larger trend, we could be entering with increasing competition. The number of companies rising, however, does bring optimism, as there is lots of competition for the job market, meaning more opportunities may be arising. Combining the 2 sets of data and analyzing the employees per company gives a much more bleak outlook, showing that employee efficiency has grown by almost double in the last 5 years. This is likely due to the AI threat to our job market increasing, and if the employees per company continues to fall, it could lead to a very difficult job market going forward.

3.5

Looking at the software publishing industry from a monetary perspective, there is still a very steady increase in annual wages as well as salary. So, although the employment numbers themselves may appear bleak, from 2024 our potential compensation is still increasing as well as the investment in our industry. There was a correction in 2022, which could lead to a concern, however, the following years, 2023 and 2024, grew steadily. Combining with figure 4, we can see why this may be the case, as the number of employees increased by around 50K.

The likelihood is there were more entry level developers entering the market, drawing the average salary closer to the entry level salary. Additionally, the total national wages also flattening could mean that there was a slight firing or retiring of senior level employees with the mass hiring in 2022 that drew the average salary down. Having a higher number of entry or lower-level employees could draw concern as we enter the market should the investment in compensation slow down.

3.6

The number of companies in each state is very informative, as the highest state is Florida, and right behind is New York. Only looking at the companies in each state would make it seem as though Florida is the most appealing to look for job opportunities, and if a new grad is looking to simply find the quickest or easiest job market, then Florida seems most appealing. However, combining this with figure 7 shows a different story where the amount of money in the state in the industry is low, meaning if there are job opportunities, they are likely smaller companies with a potentially lower compensation or lower number of employees.

3.7

The monetary geographical flow shows a vastly different story, where the most amount of employee wages are spent in California and Washington. Although the number of companies in these 2 states is slightly lower, as shown in figure 6. It is likely that larger companies like Amazon and Microsoft are drastically increasing the number of wages in these states. Meaning that the opportunities could be greater for a smaller number of companies, but it also means that we would likely be working for a medium to large company. If we want to work for a medium to large-sized company, it is more likely that there are more opportunities in Washington and California to get a job with such companies.

3.8

This chart reveals that analyzing base salary alone severely underestimates the true gender wealth gap. In the lower paying "Service Roles," the connecting lines are short, indicating that total compensation barely exceeds the base pay. However, in "Management and Tech Roles,"

the line is massive. Another thing is the initial gap between men and women at the base salary level (hollow dots). This proves that high paying sectors don't just offer higher wages but accelerate the wealth divide by their compounding benefits.

3.9

This chart illustrates a huge contrast in compensation models. Private industry relies on a cash heavy model, which means a dominated base wage component. In contrast, the public sector (state & local government) dedicates a significantly larger, thicker proportion of its total compensation package to stable, wealth-building benefits specifically towards retirement, savings, and insurance. This suggests that while private industries may offer a higher ceiling for base pay, the public sector acts as a powerful economic equalizer by offering a much higher, guaranteed "benefits floor."

3.10

The data exposes a structural disadvantage in the workforce that drives the wealth gap. The left panel shows a relatively balanced landscape for high-benefit professional roles, with women holding a slight advantage. However, the right panel reveals the true economic divide: specifically, how women disproportionately carry the burden of the low benefit economy. Women outnumber men by over 6 million in service and office roles where employer provided wealth building benefits are practically nonexistent.

3.11

This chart illustrates how, although organized labor raises the baseline compensation for every demographic, the percentage increase is not evenly distributed among them. Historically disadvantaged groups see the most dramatic financial benefits compared to their nonunion counterparts. This indicates that unions serve as a disproportionately powerful economic shield for demographics most affected by the broader wealth gap.

3.12

This graph shows the AI adoption rate as a time series between 2017 and 2025. In the beginning of the graph, in 2017, the AI adoption rate is low, at around 20%. Artificial intelligence has always been a field of study long before this period but has still yet to make any real headway in the industry. It climbs steadily. However, after 2022, AI adoption surges dramatically, going from 50% to 88%. This can be attributed to the rise of LLMs and generative AI such as ChatGPT, Gemini, and Co-Pilot, which have been utilized extensively for both personal, professional, and academic use cases. Overall, this graph clearly shows that AI has made its way into all the workflows of major industries across the globe. Whether the effectiveness of AI for productivity is marginal or not remains a constantly evolving question, however.

3.13

This graph shows the unemployment rate for Computer Science Major-Related Graduates over a time series, between 2017-2015 (same time frame as in 3.12). In the beginning of 2017, unemployment rate remains fairly low at around 3.2%, well below the national average of that year of 4.4%. It remains this way until about the 2020-2021 timeframe, where we see a surge in unemployment rising up to 4.9%. However, this is mainly attributed to the COVID-19 Pandemic, where unemployment surged in all fields, not just in ones related to Computer Science. In 2023, unemployment recovers, dropping to 3.0%, which can be attributed to the recovering economy post COVID-19. However, moving past that, we see in 2025, unemployment surges to nearly 6.4%. This number is well above the national average and ranks amongst the highest for graduates across all majors. People in the computer science community are especially concerned by this, thinking it could be attributed to artificial intelligence replacing humans in the field. While it is too soon to claim this conclusively, it does raise important questions about how AI/AI Automation might be reshaping the labor market. Is this temporary? Is artificial intelligence Automating jobs entirely? Or is the productivity boost just so great that companies don't need as many employees? These are interesting questions to think about, but at the present there aren't enough data points for a definitive answer.

3.14

This horizontal bar chart shows Computer Science Occupation Unemployment Rates for the year 2025. It limits to the roles: Web Developers, Web and Digital Interface Designers, Software QAs (Quality Assurance) roles, and Software Developers. We see that web developers have the highest rate of unemployment, at 10.6%. Web and Digital Interface Designers follow at 7.8%, Software QAs at 7.1%, other Computer Occupation roles at 3.4%, and Software Developers at 3.3%. Relating back to the analysis done in 3.13, this graph suggests that certain roles within Computer Science may be more vulnerable to the labor market shifts influenced by AI/AI Automation than others.

6. Conclusions

Geographically, Washington and California have the best job prospects based on the number of companies and the salaries they pay. The number of companies has grown, but the average number of employees per company has gone down. **Threats by AI:** Given a recent surge in AI adoption rate, research shows that jobs that are repetitive and clerical face the highest risk of automation by AI. Compensation/Wage Trend for Employment: Current wage trends reveal a significant gender gap in 'hidden' compensation, particularly regarding employer-provided benefits. AI Adoption Increase vs. Unemployment: It is too early to conclude that AI will take over computer science-related occupations, but it raises questions about the direction the job market is headed in.